MUSIC DECODER · VISUAL SPEC FOR CURSOR

Theory map

A new Learn feature that explains every music term, shows how the terms connect, and plays an example of each one. It includes a narrated guided tour that runs like a video.

How to use this document

Each page is a screen from the working prototype (`reference-prototype.html`) with numbered callouts. The notes say what to build and how it behaves. Match the layout, states and behaviour. Use the app's existing components, buttons and fonts where they already exist. The lane colours and the amber highlight are new tokens (page 2).

Files that come with it

- `cursor-prompt.md` : the build instructions
- `theory-content.ts` : all terms, examples, map layout and tour steps, typed and tested
- `content.test.ts` : integrity tests for that file
- `reference-prototype.html` : open in a browser to see and hear the behaviour

Contents

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8. Principles
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10. Glossary
11. Phone layout
12. Dark theme
13. Acceptance checklist

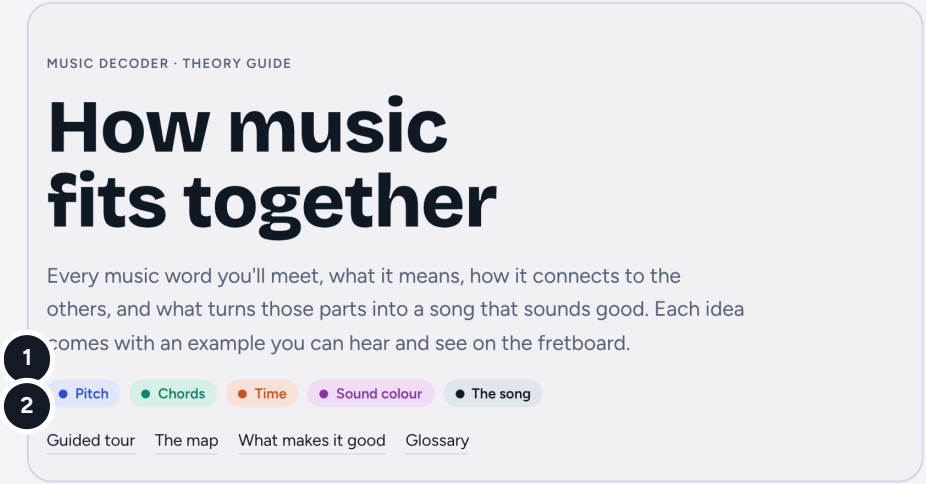
Prototype screenshots, 24 October 2026

1 · STRUCTURE AND TOKENS

One page, six sections, top to bottom

Route suggestion: `/learn/theory`, linked from the Learn page and the main menu. Everything is client-side; no backend changes.

Hero	Guided tour	Map + detail	What makes it good	Six steps	Glossary
title, lane legend, jump links	narrated player, 26 steps	SVG map, panel for the selected term	5 principles, compare A/B	recipe + 9-bar mini song	table of all 19 terms



Hero: (1) lane legend chips, (2) in-page jump links.

Lane colours (new tokens)

LANE	LIGHT COLOUR	LIGHT TINT	DARK COLOUR	DARK TINT	TERMS
Pitch	#2E4FD6	#E3E9FD	#8299FF	#1E2850	note, interval, scale, key
Chords	#0B8670	#D8F1EA	#3FC9A7	#0F322B	chord, progression, cadence
Time	#C9511F	#FBE4D9	#FF8C5C	#3C2116	beat, tempo, meter, rhythm
Sound colour	#8B33A5	#F1E0F7	#D27DEC	#341C3D	texture, timbre, dynamics
The song	#131925	#E2E7EF	#E7EBF3	#252D3A	melody, harmony, groove, arrangement, form, principles

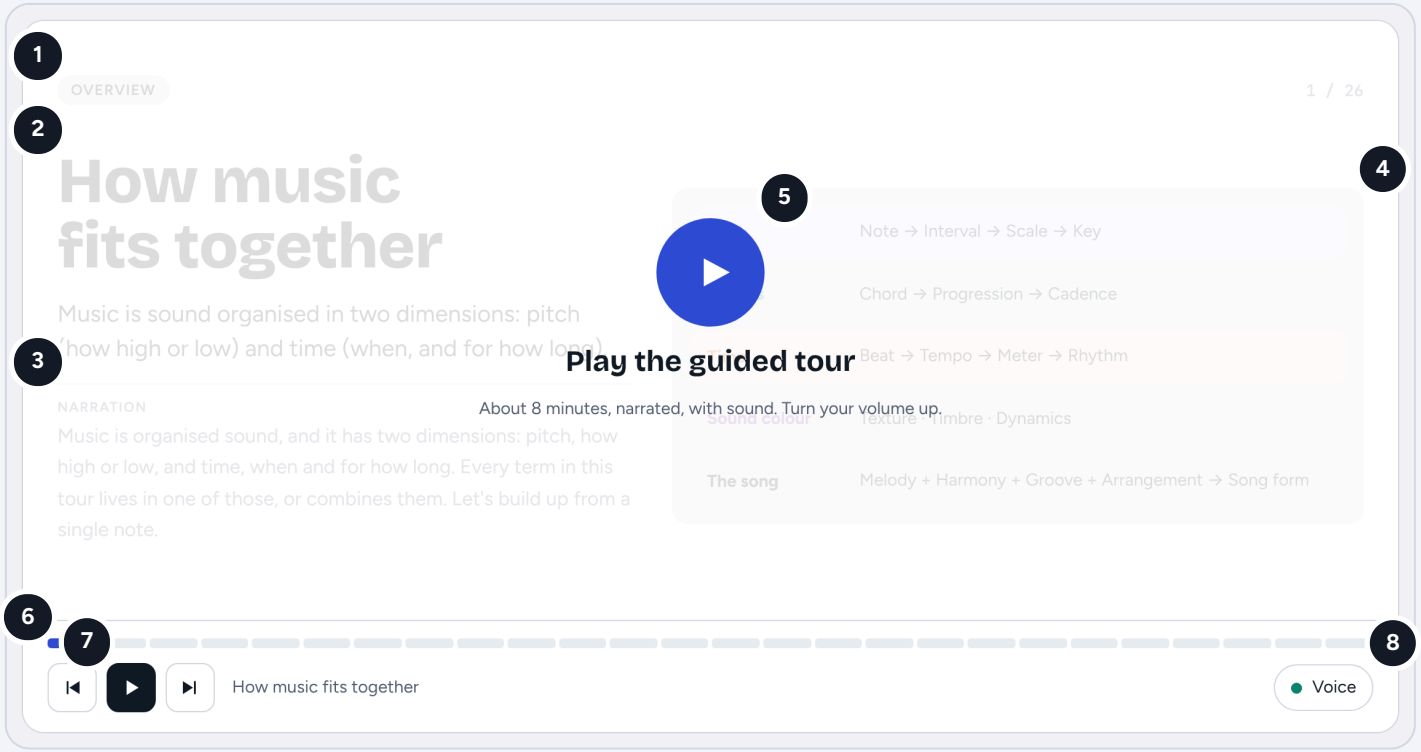
Other tokens

Highlight	#F2B233 light · #F5BE45 dark, text #2A1C00	anything sounding right now: fret dot, grid cell, block
Accent	use the app's primary colour	tour play button, progress
Type	Display (prototype: Bricolage Grotesque 700–800) for titles; body (Figtree); mono (JetBrains Mono) for numerals, BPM, fret numbers. Swap in the app's fonts if it has a set.	
Shape	Cards 18px radius, 1px border; chips fully rounded; visual wells 12px radius on the page background.	

2 · GUIDED TOUR

Idle: the player before the first press

The tour is the "video". It sits at the top of the page and opens on step 1 with a start overlay.

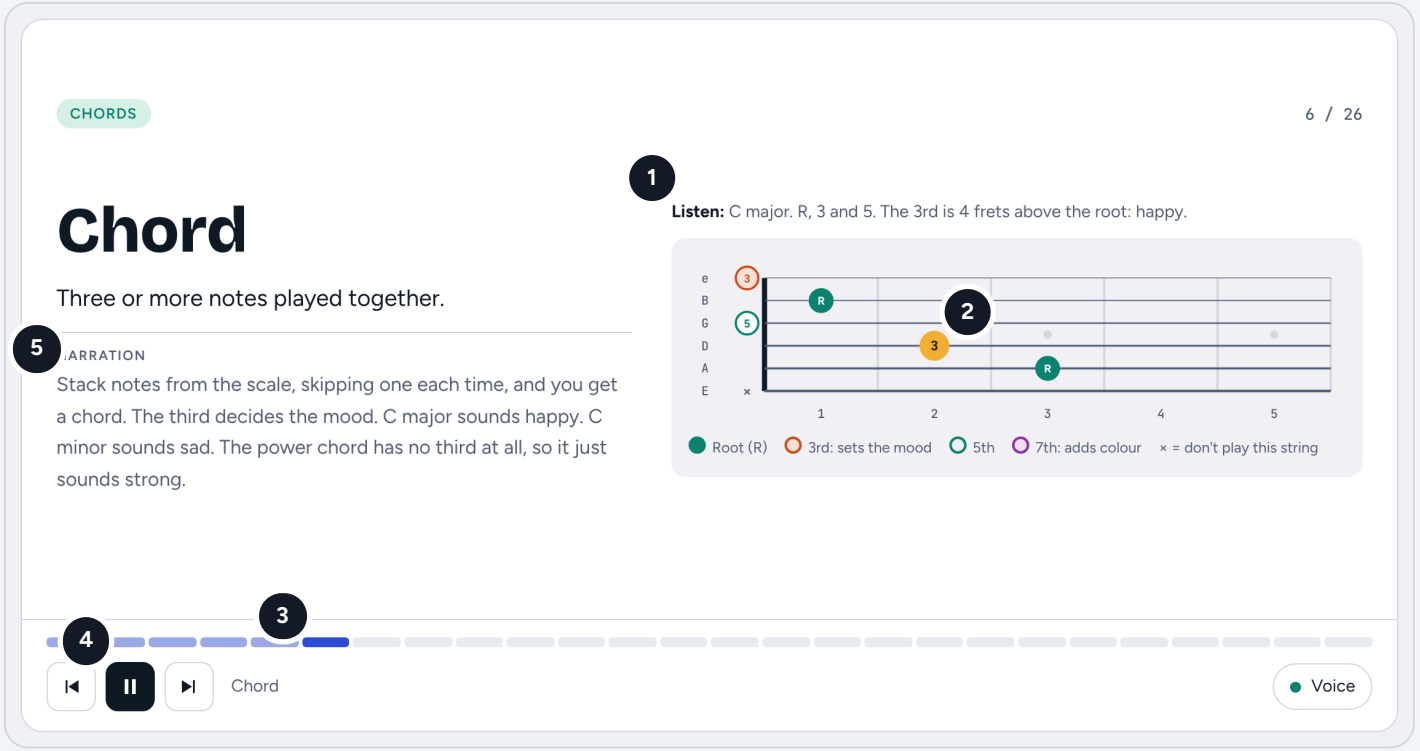


1. **Lane chip**
   Eyebrow for the step, coloured by its lane (Overview, Pitch, Chords, …, What makes it good · 1 of 5, Finale).
2. **Title and definition**
   `TourStep.title` and `definition`. Large display type.
3. **Narration**
   The exact text being spoken. Doubles as subtitles when the voice is off. `aria-live="polite"`.
4. **Visual stage**
   Shows the current example's visual. Step 1 shows the five-lane overview list.
5. **Start overlay**
   Covers the player until the first press. This click must create/resume the audio context.
6. **Progress**
   One segment per step (26). Done = tinted, current = accent. Clicking jumps to that step.
7. **Transport**
   Previous, play/pause, next, then the current step name.
8. **Voice toggle**
   Uses the app's existing speech helper and setting. Disabled with a tooltip when speech is unavailable; subtitles still advance on a timer.

3 · GUIDED TOUR

Playing: narration, then the examples, in sync

Each step runs narrate → 300 ms → every example in `tour[]` with 700 ms gaps → next step.



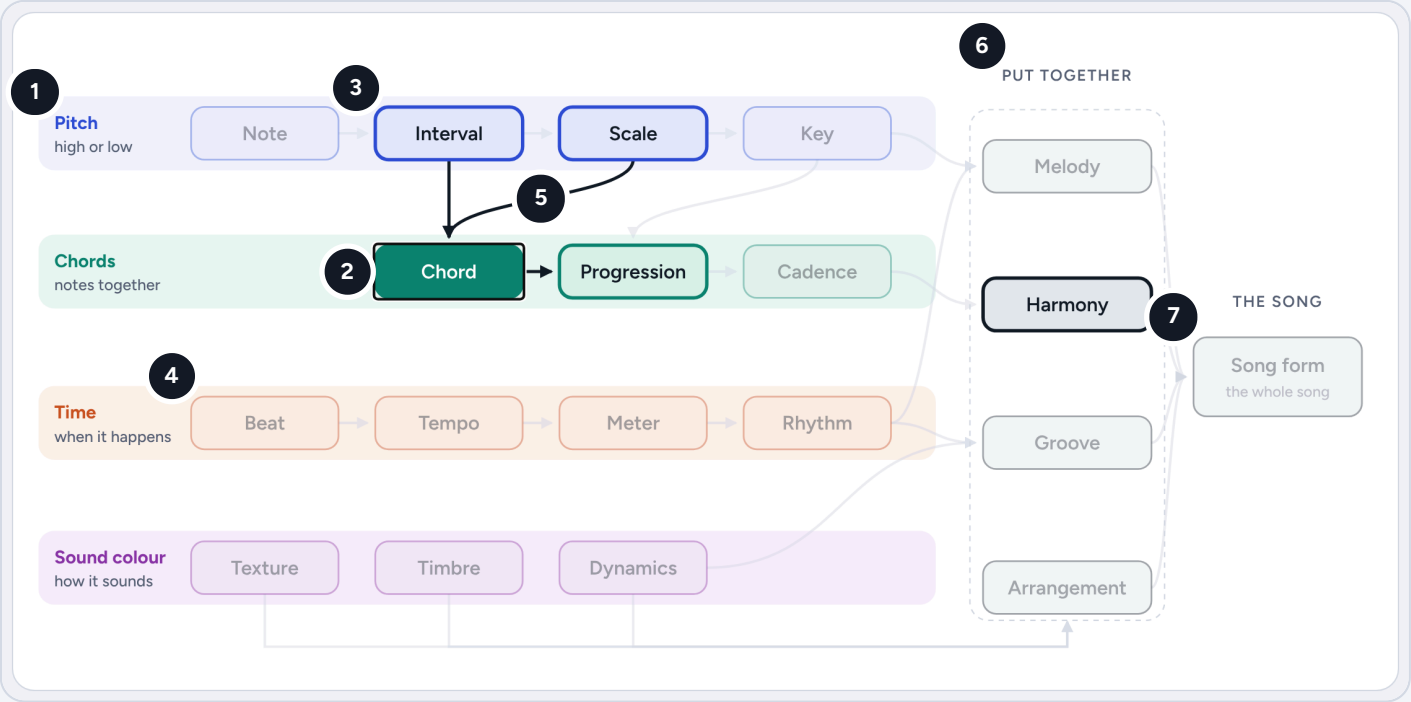
1. **Listen line**
   "Listen: <label>. <sub>" names the example that is playing and what to hear in it.
2. **Live highlight**
   The sounding note lights amber and scales to 1.2×, in sync with the audio (no scaling with reduced motion).
3. **Current step**
   Progress segment for the step that is playing.
4. **Pause**
   Replaces Play while running. Pause cancels speech and audio; Play restarts the current step from its narration.
5. **Subtitles**
   Narration text stays visible while the examples play.

The tour also selects the step's term on the map below, so the map and detail panel follow along. Any manual action elsewhere on the page (selecting a node, pressing an example) pauses the tour.

4 · CONCEPT MAP

Four lanes that meet in the song

An SVG map (viewBox 1030 × 492) built from `MAP_NODES`, `MAP_EDGES` and `edgePath()`. Chord is selected here.



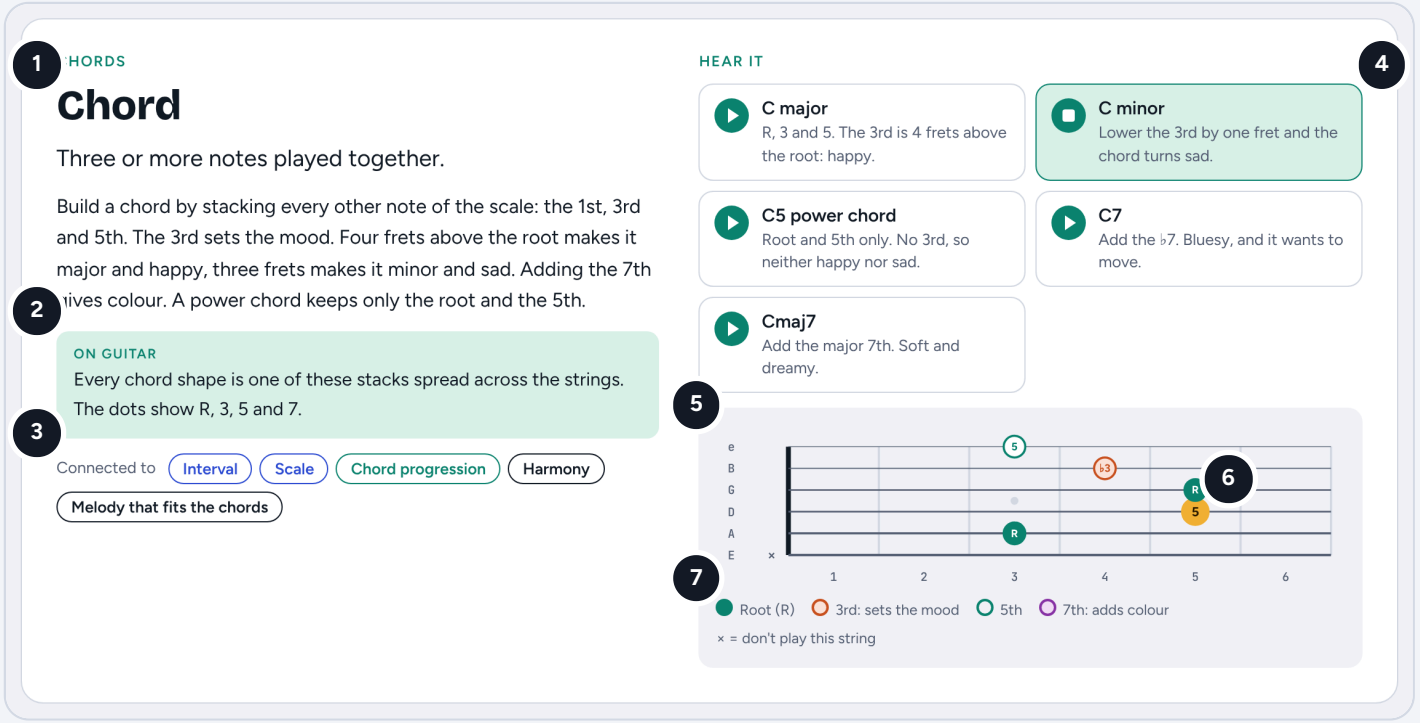
1. **Lane band**
   Tinted row with lane name and a two-word hint.
2. **Selected node**
   Filled with the lane colour, text in the surface colour.
3. **Related nodes**
   Thicker outline. Related = `connections(id)`.
4. **Dimmed**
   Everything unrelated drops to about 40% opacity while a node is selected.
5. **Hot edges**
   Edges touching the selection turn ink-coloured with a dark arrowhead and are drawn on top.
6. **Put together column**
   Melody, harmony, groove, arrangement in a dashed frame.
7. **Song form**
   The end point of the whole map.

Nodes are focusable buttons (Enter or Space selects). On narrow screens the SVG keeps a min width of 880px and scrolls inside its card.

5 · TERM DETAIL

The panel under the map

Shows the selected term. Left: explanation. Right: examples you can play, and the visual for the last one pressed.

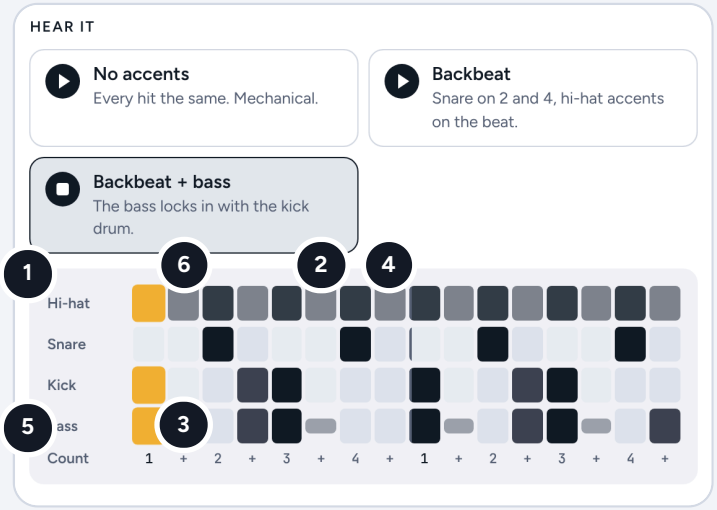


1. **Heading**
   Lane eyebrow, name, one-line definition (larger), then the body text.
2. **On guitar**
   Practical tip in a lane-tinted box.
3. **Connected to**
   Chips for every connection. Clicking one selects it, updating the map and the panel.
4. **Example buttons**
   Play icon turns into a stop square while playing; a second press stops. Only one example plays at a time, app-wide.
5. **Fretboard**
   Standard tuning, high E at the top (TAB order), nut when fret 0 is shown, fret numbers below, inlays at 3/5/7/9/12, × for muted strings. Follow the left-handed setting. Reuse the app's Fretboard component if it can show static dots.
6. **Dot roles**
   Root filled; 3rd orange; 7th and blue note purple; others outlined. Chord dots show R/3/5/♭7, other dots show note names. Amber = sounding now.
7. **Legend**
   Explains the dot colours for this example (`legend`: default, chord or blue).

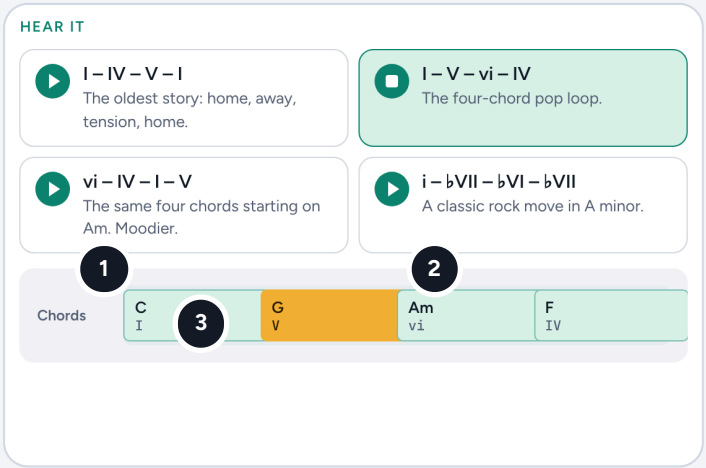
6 · EXAMPLE VISUALS

Grid and blocks

Three visual types cover every example: `fret`, `grid` and `blocks`. Each element carries the highlight id used by the events.



Grid visual (Groove, Backbeat + bass)



Block visual (Progression, I – V – vi – IV)

1. **Row names**
   Instrument per row: Hi-hat, Snare, Kick, Bass, Guitar, Click, Beat, Tune.
2. **Hit cell**
   Lane colour mixed by velocity: 25% + 75% × velocity. Rests stay empty.
3. **Tie**
   A thin bar continues a held note; it lights together with its head cell.
4. **Bar line**
   Every `stepsPerBar` cells. Alternate beats are shaded slightly.
5. **Count row**
   1 + 2 + … from `stepsPerBeat`; beat 1 of each bar in ink.
6. **Lit cell**
   Amber while sounding.

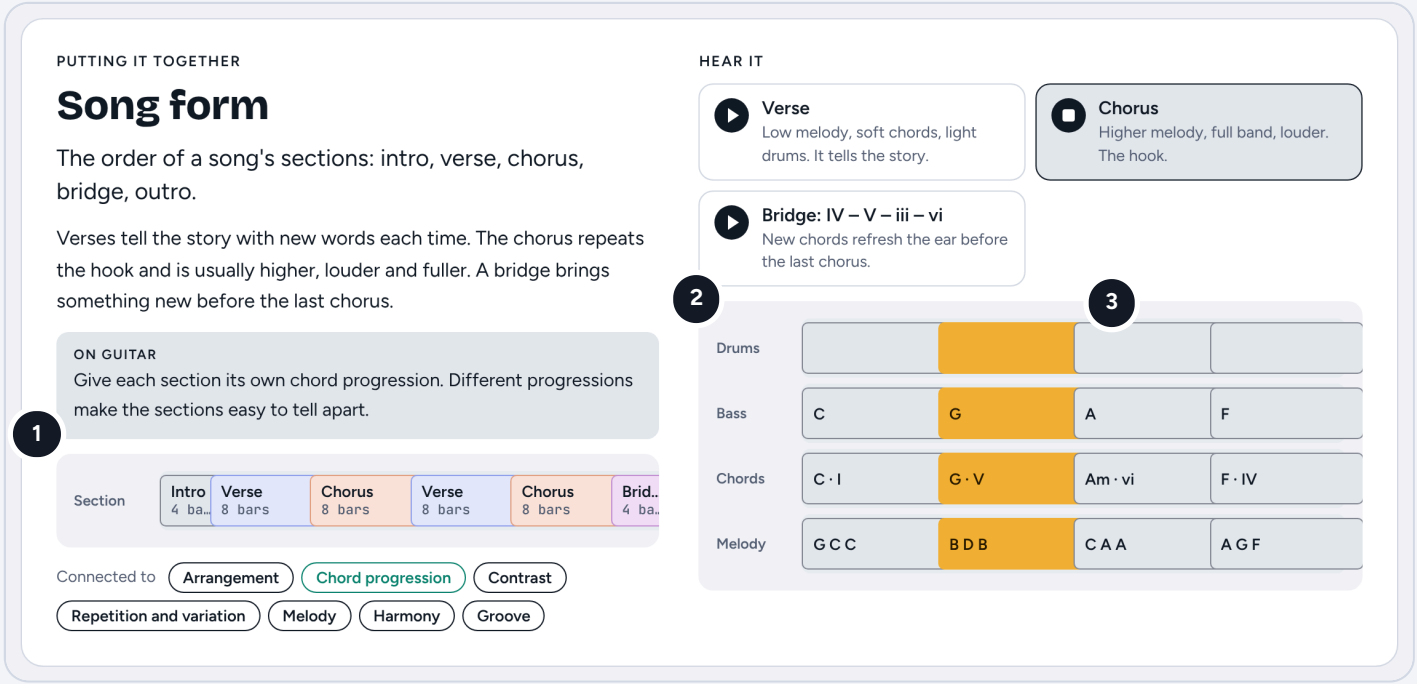
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1. **Block**
   One per chord or bar. Width is proportional to beats. Name in bold, Roman numeral or detail in mono.
2. **Lit block**
   Amber for the whole time its chord sounds.
3. **Numeral**
   Upper case = major, lower case = minor.

7 · SONG FORM

Static figure plus layered examples

Song form adds a static timeline figure to the left column. Its examples use the layered block view (one row per instrument, one block per bar).

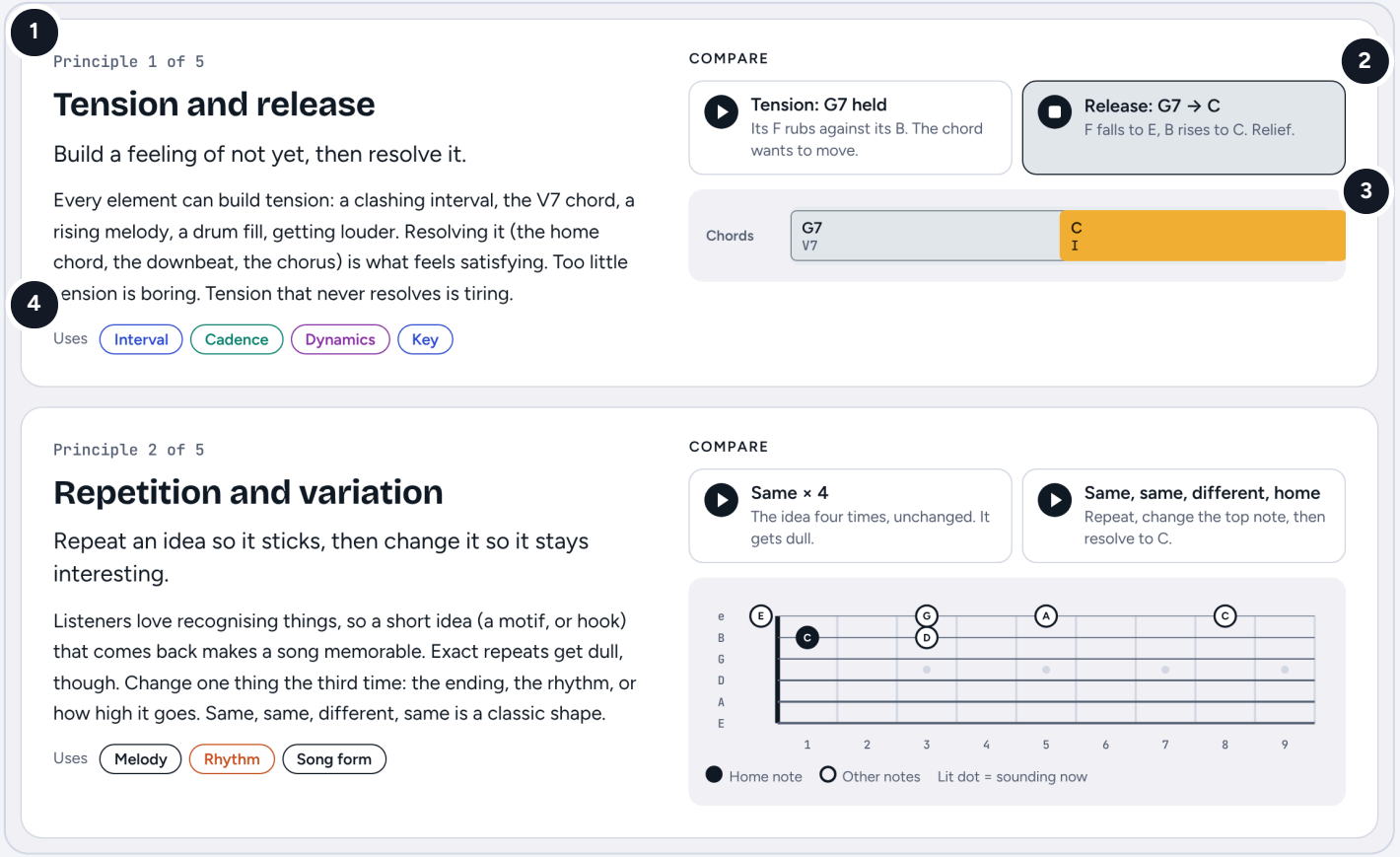


1. **Form timeline**
   Intro, Verse, Chorus, Verse, Chorus, Bridge, Chorus, Outro, coloured by section type. Scrolls sideways inside its well on small screens.
2. **Layer rows**
   Drums, Bass, Chords (chord · numeral), Melody (note names). A layer with no block in a bar is silent in that bar.
3. **Lit bar**
   Every layer playing in the current bar lights at once.

8 · WHAT MAKES IT GOOD

Five principles, each a comparison

Same panel pattern as the term detail, in a vertical list. Example data: `PRINCIPLES`.

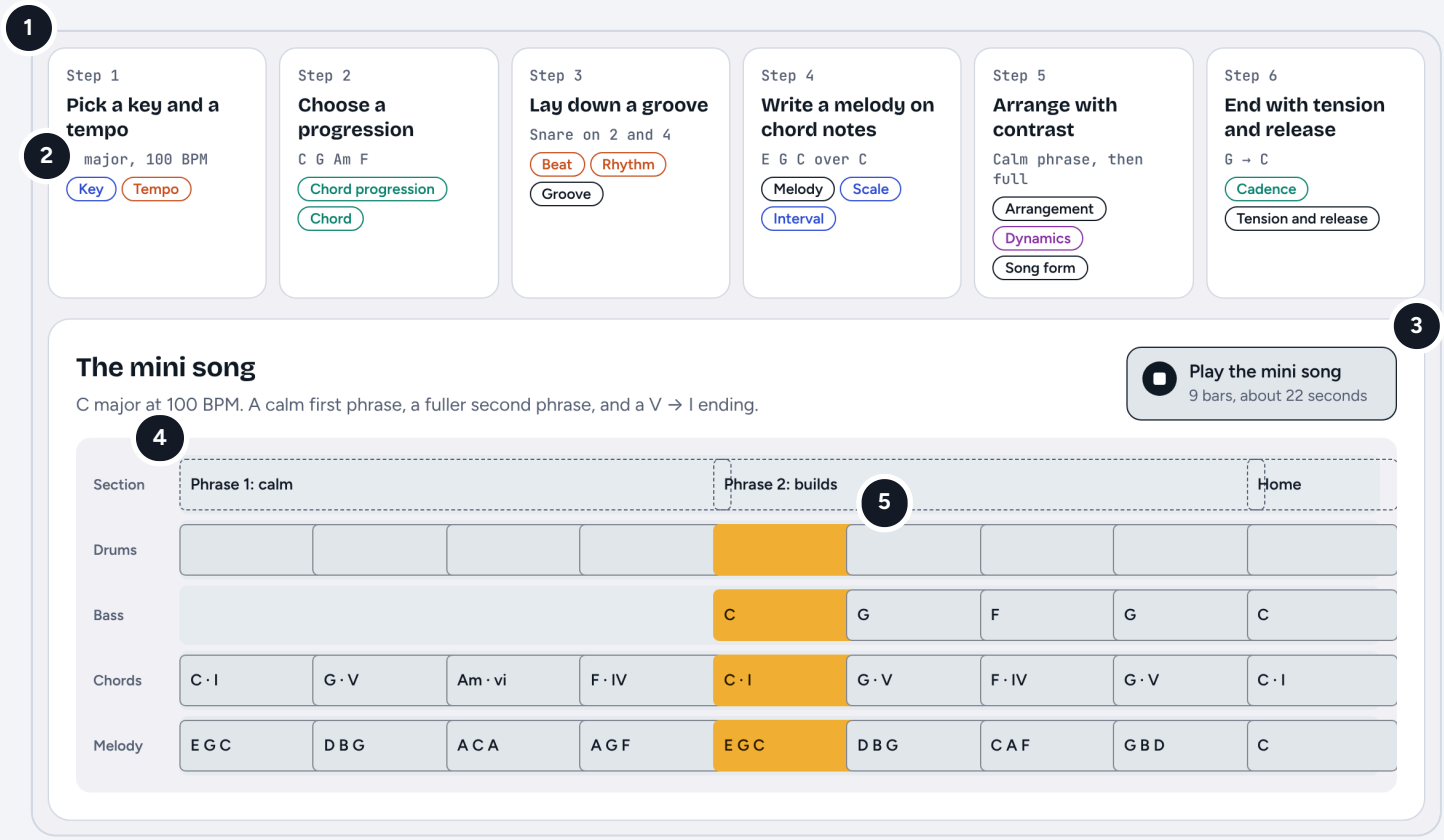


1. **Counter**
   "Principle n of 5" in mono, then name, one-liner and body.
2. **Compare buttons**
   Two versions (A and B) with the same play and stop behaviour as other examples.
3. **Live visual**
   The visual for the last-pressed version, lit while playing.
4. **Uses**
   Chips for the related terms. Clicking selects the term and scrolls to the map.

9 · PUTTING IT TOGETHER

Six steps and the mini song

A real sequence, so numbered. The mini song (`MINI_SONG`, 9 bars at 100 BPM) uses every step.



1. **Step card**
   Step number, title, mono example, term chips. Six columns on desktop, two on phone.
2. **Term chip**
   Selects that term on the map and scrolls to it.
3. **Play button**
   Plays the mini song (about 22 seconds); press again to stop.
4. **Section row**
   Dashed labels: Phrase 1: calm, Phrase 2: builds, Home.
5. **Lit bar**
   Each layer's block lights for the bar that is playing. Bass enters in bar 5.

10 · GLOSSARY

Every term at a glance

Generated from `TERMS`. Table scrolls sideways on phones.

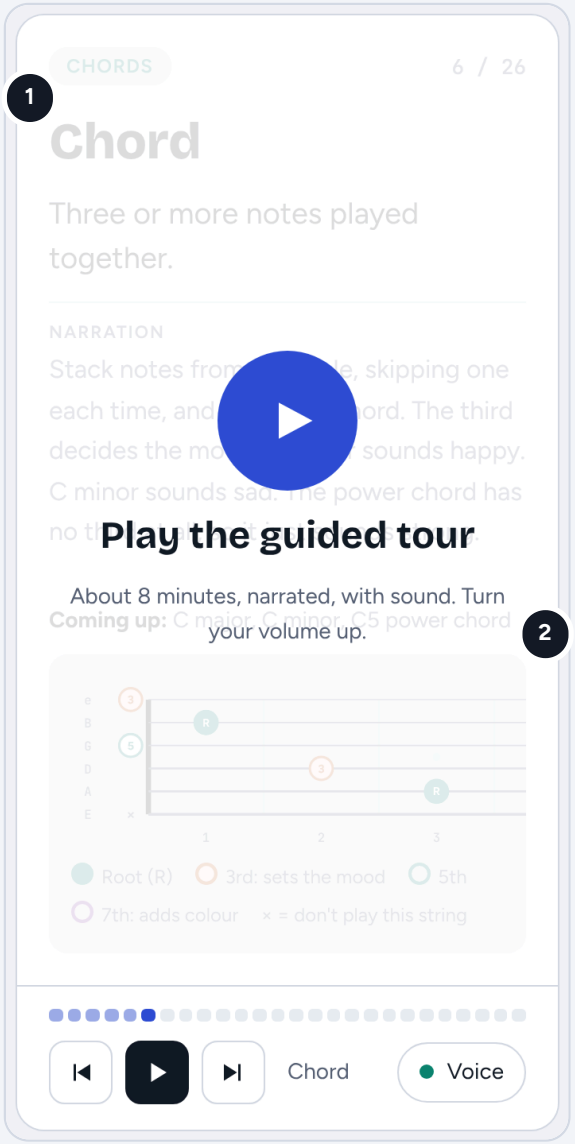
1 TERM	WHAT IT IS	2 EXAMPLE	3 CONNECTS TO
Note	One musical sound with a fixed pitch and a name.	`C, then C again at fret 12`	Interval, Scale, Melody
Interval	The distance between two notes, counted in semitones. On guitar: frets.	`C → E = major 3rd, 4 frets`	Note, Chord, Scale, Tension and release
Scale	A ladder of notes chosen from the 12, played in order.	`C D E F G A B`	Interval, Key, Chord, Melody, Note
Key	The home note plus its scale. It tells you which notes and chords belong.	`C major: home is C`	Scale, Chord progression, Melody, Cadence, Tension and release
Chord	Three or more notes played together.	`C = C E G`	Interval, Scale, Chord progression, Harmony, Melody that fits the chords
Chord progression	A sequence of chords that repeats under a song.	`C G Am F (I-V-vi-IV)`	Key, Chord, Cadence, Harmony, Song form
Cadence	The chord move that ends a phrase. Music's punctuation.	`G → C, a full stop`	Chord progression, Key, Tension and release, Harmony
Beat	The steady pulse you tap your foot to.	`Foot taps at 90 BPM`	Tempo, Meter, Rhythm, Space and timing
Tempo	How fast the beat goes, in beats per minute (BPM).	`70 ballad, 120 rock, 170 punk`	Beat, Groove, Meter

1. **Term**
   Lane-coloured dot and name. Clicking selects it on the map and scrolls up.
2. **Example**
   `glossaryExample` in mono.
3. **Connects to**
   Names from `connections(id)`.

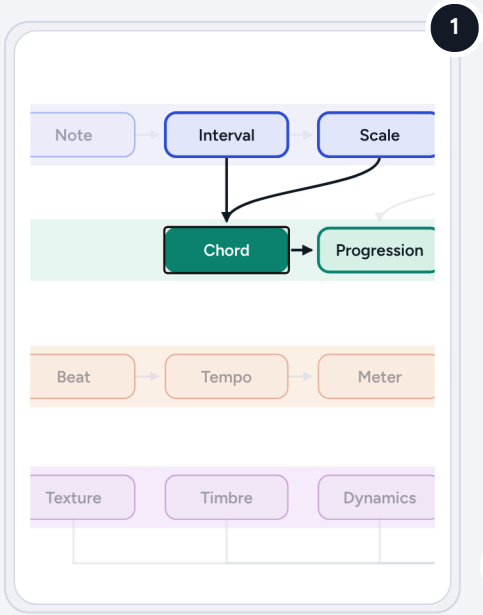
11 · PHONE (390 PX)

Everything stacks; only the map and wide visuals scroll sideways

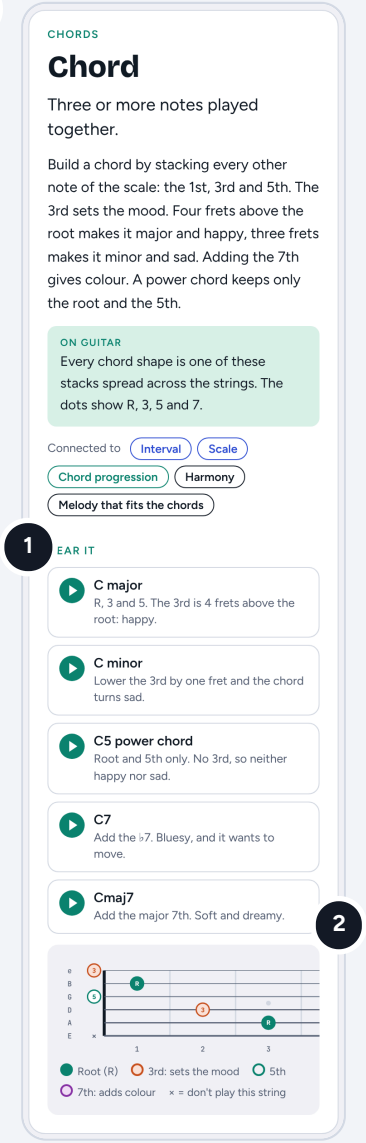
The page itself never scrolls horizontally. Keep 16 px side gutters plus safe-area insets.



Tour



Map (scrolls sideways)



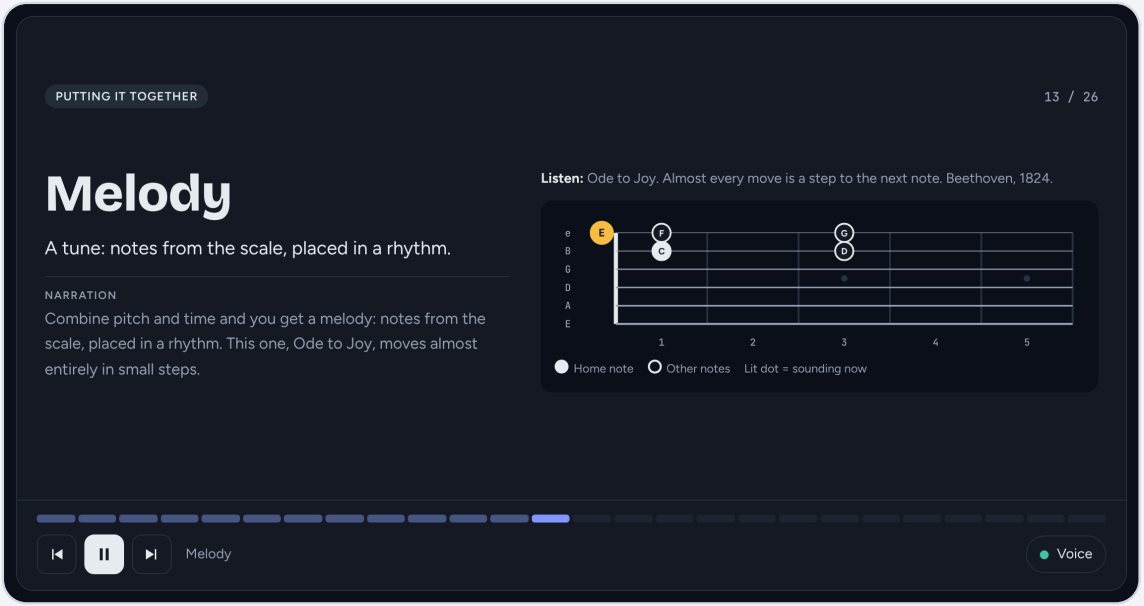
Detail, stacked

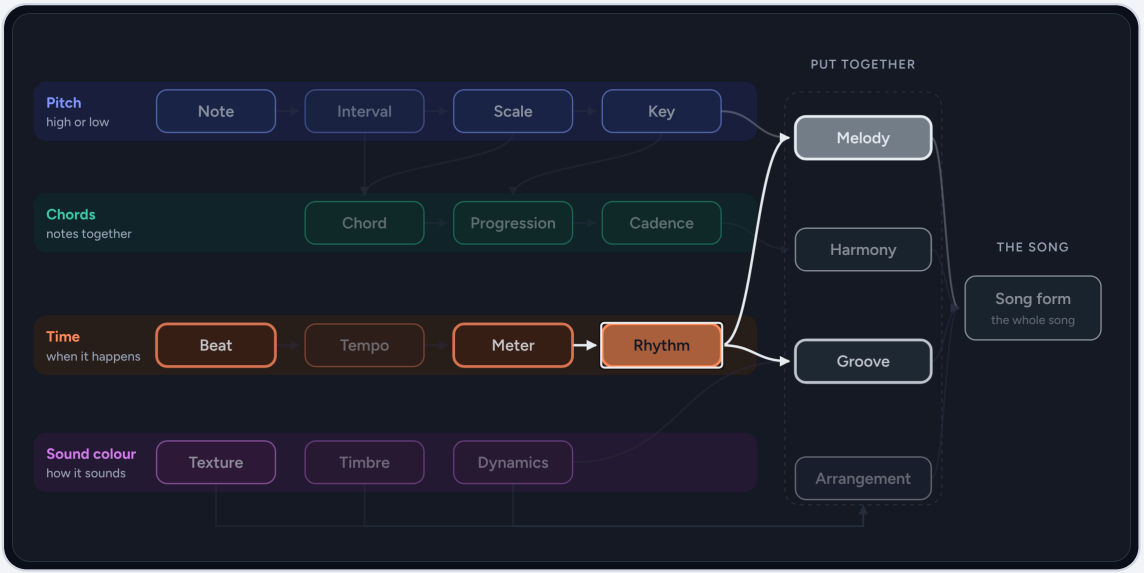
1. **Tour**
   Text above the visual. Overlay and controls keep the same order.
2. **Map**
   Horizontal scroll inside the card; selection still drives the panel below.
3. **Detail**
   Text, then example buttons (full width), then the visual.

12 · DARK THEME

Same layout, lighter lane colours

If the app has a dark theme, define each lane token for both themes (values on page 2). The amber highlight stays amber.





1. **Surfaces**
   Use the app's dark surface and border tokens.
2. **Lanes**
   Lighter lane colours and deep tints so labels keep contrast.
3. **Selected node**
   Lane colour fill with dark text.

13 · ACCEPTANCE

Done when

The same checklist is in `cursor-prompt.md`.

1. Route renders without errors; no horizontal page scroll at 390 px.
2. Selecting any of the 19 map nodes updates the map highlight and the detail panel.
3. Every example plays through the app's audio engine, lights its visual in sync, and stops cleanly when pressed again or when another example starts.
4. Only one thing plays at a time across the page (tour, examples, mini song).
5. Tour: start overlay → 26 steps in order; pause, resume, previous, next and clicking a progress segment all work; the tour selects each term on the map.
6. Narration uses the app's speech helper and voice setting; with voice off, subtitles advance on a timer.
7. Fretboard respects the left-handed setting if the app has one.
8. Reduced motion: no scaling or sliding; highlights still change colour.
9. Keyboard: map nodes, chips, example buttons and tour controls are reachable and visibly focused.
10. Leaving the page stops audio and speech.
11. Unit tests: content integrity (content.test.ts) and the player's stop/cancel logic. Browser tests: select a node, play an example, run and pause the tour, phone layout.